

PROJECT REPORT

Operations Analytics-1: Time-Series Forecasting Report

Target Variable: Weekly Cold Coffee Demand

Methodology: ARIMA Pipeline (Stationarity, AIC/BIC, Residual Diagnostics)

Raw Weekly Cold Coffee Demand



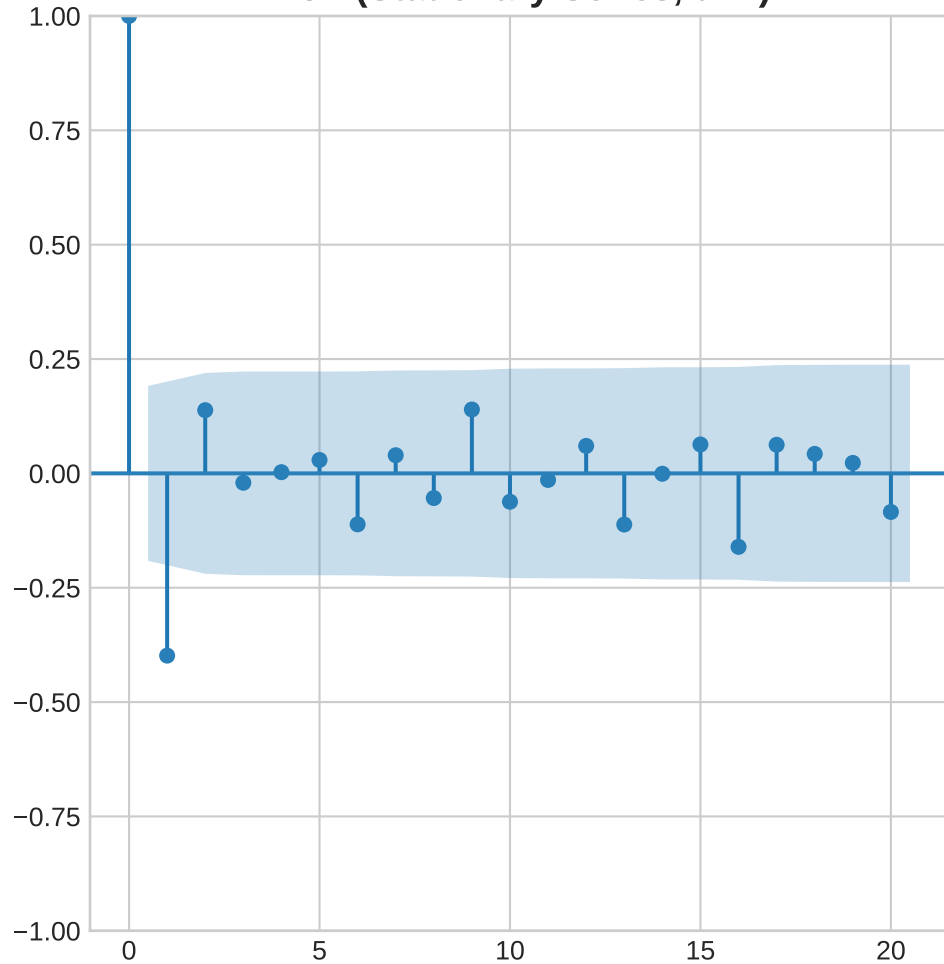
Demand with Trend (Rolling Mean)



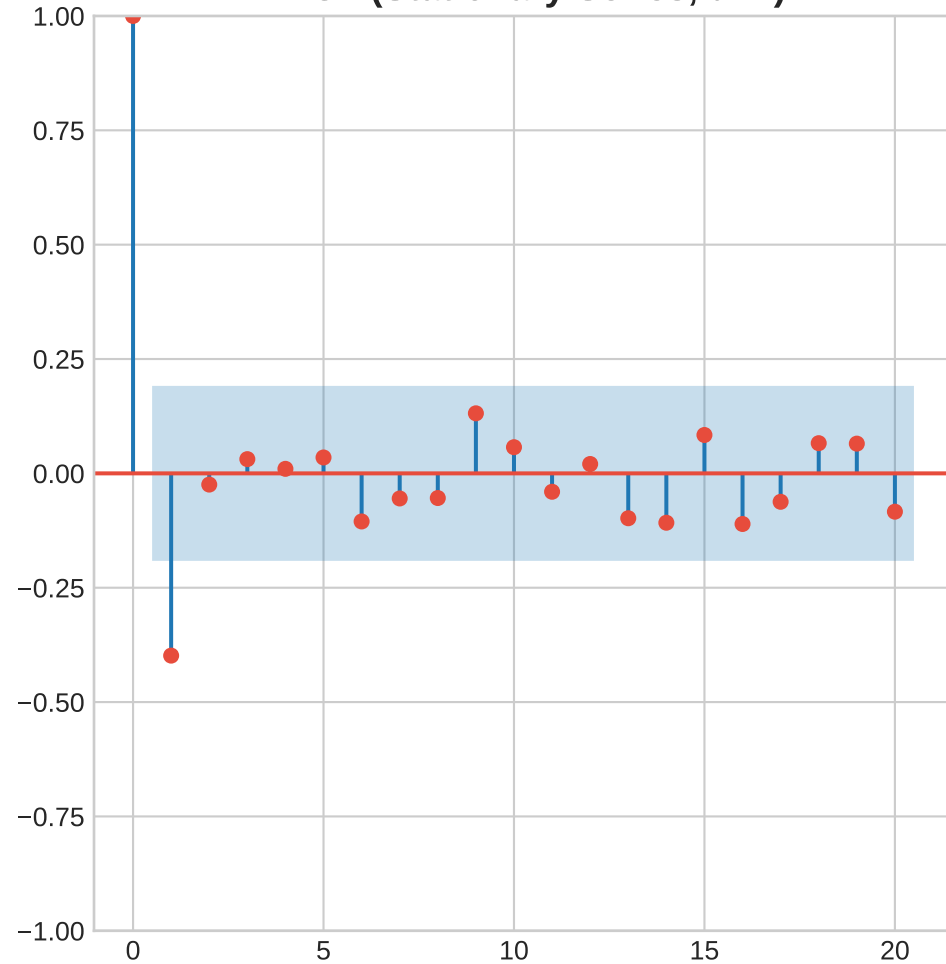
Stationarity Tests (ADF & KPSS) for Differencing (d)

d	ADF p-value	KPSS p-value	Stationary?
0	0.1856	0.1	False
1	0.0	0.1	True
2	0.0	0.1	True

ACF (Stationary Series, d=1)



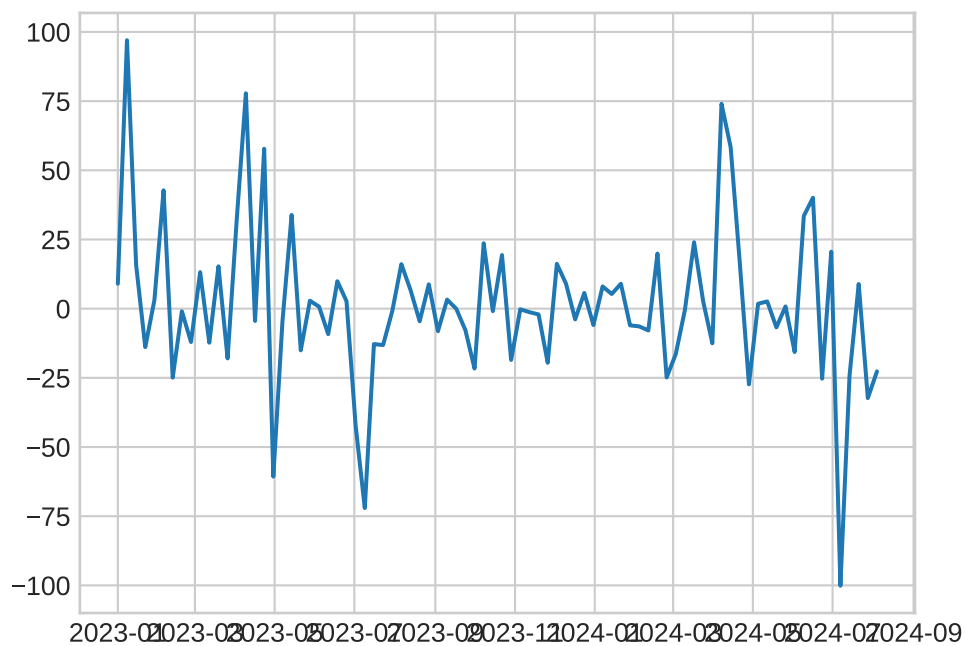
PACF (Stationary Series, d=1)



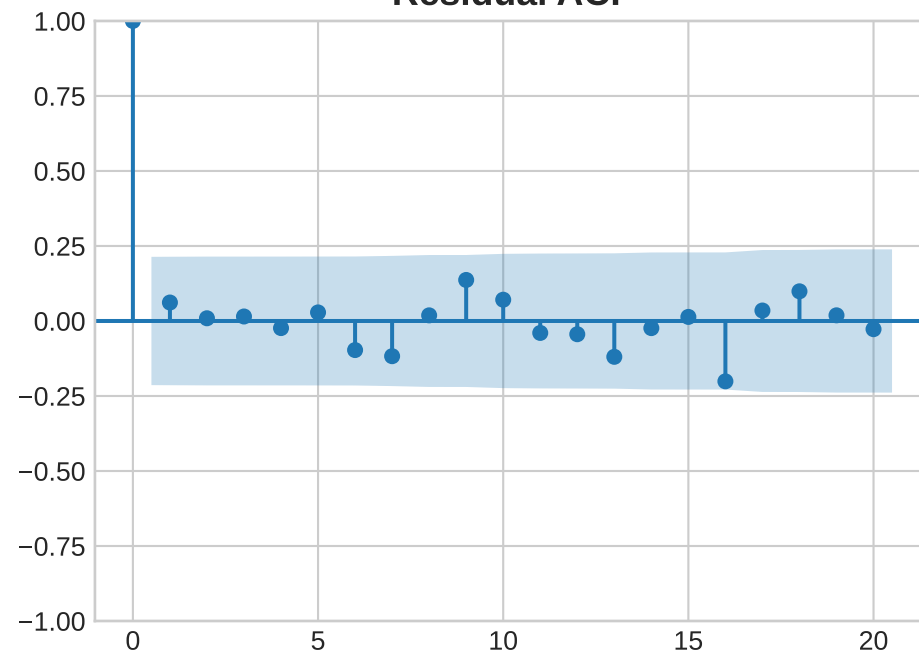
Candidate ARIMA Models: Evaluation Metrics

Model	AIC	BIC	MAE	RMSE	MAPE	Ljung_Box_P	Rank
ARIMA(1, 1, 0)	794.408	799.246	16.85	22.32	36.038	0.893	3.0
ARIMA(0, 1, 1)	797.182	802.02	15.029	21.357	36.186	0.871	6.0
ARIMA(2, 1, 0)	796.404	803.661	16.925	22.383	36.041	0.895	6.0
ARIMA(1, 1, 1)	796.405	803.662	16.91	22.37	36.041	0.894	6.0
ARIMA(0, 1, 2)	796.57	803.827	17.755	22.936	36.148	0.836	11.0
ARIMA(1, 1, 2)	798.268	807.943	17.495	22.823	36.109	0.869	12.0
ARIMA(2, 1, 1)	798.385	808.061	17.169	22.583	36.042	0.891	12.0
ARIMA(0, 1, 0)	808.275	810.694	18.136	23.305	36.406	0.069	17.0
ARIMA(2, 1, 2)	800.132	812.226	18.168	23.342	36.462	0.878	17.0

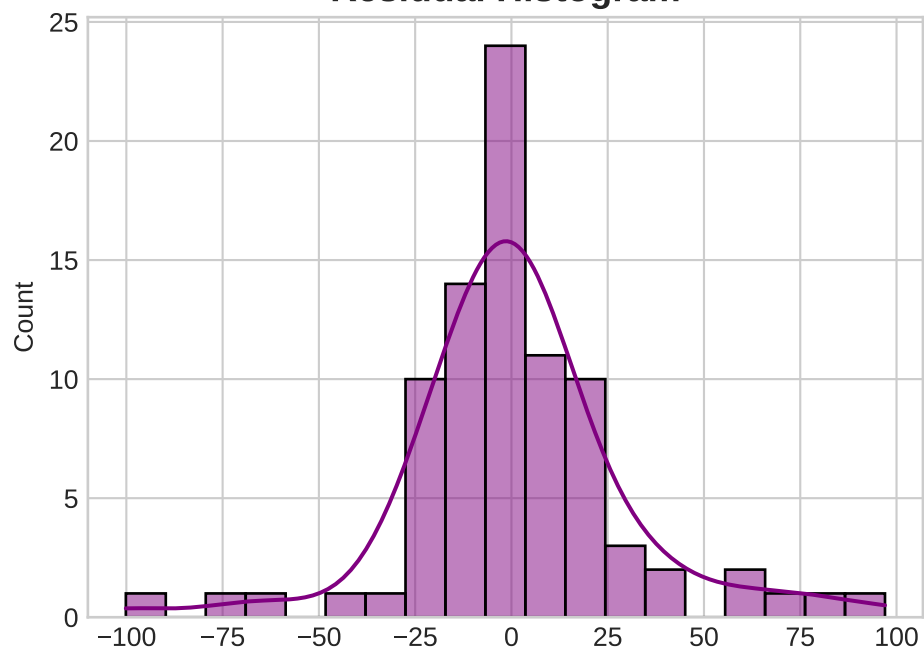
Residual Time Series



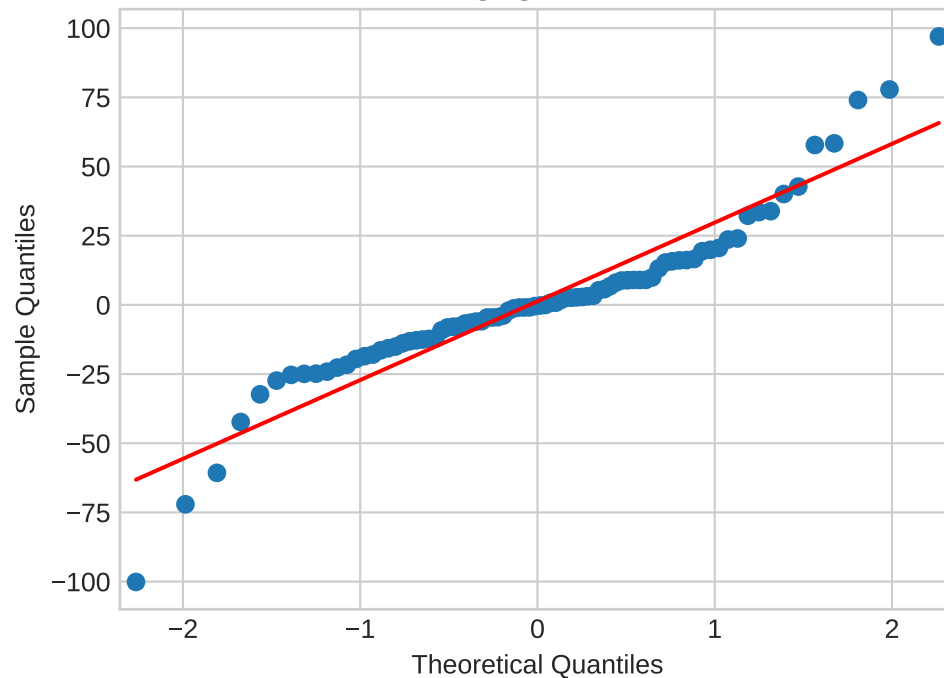
Residual ACF



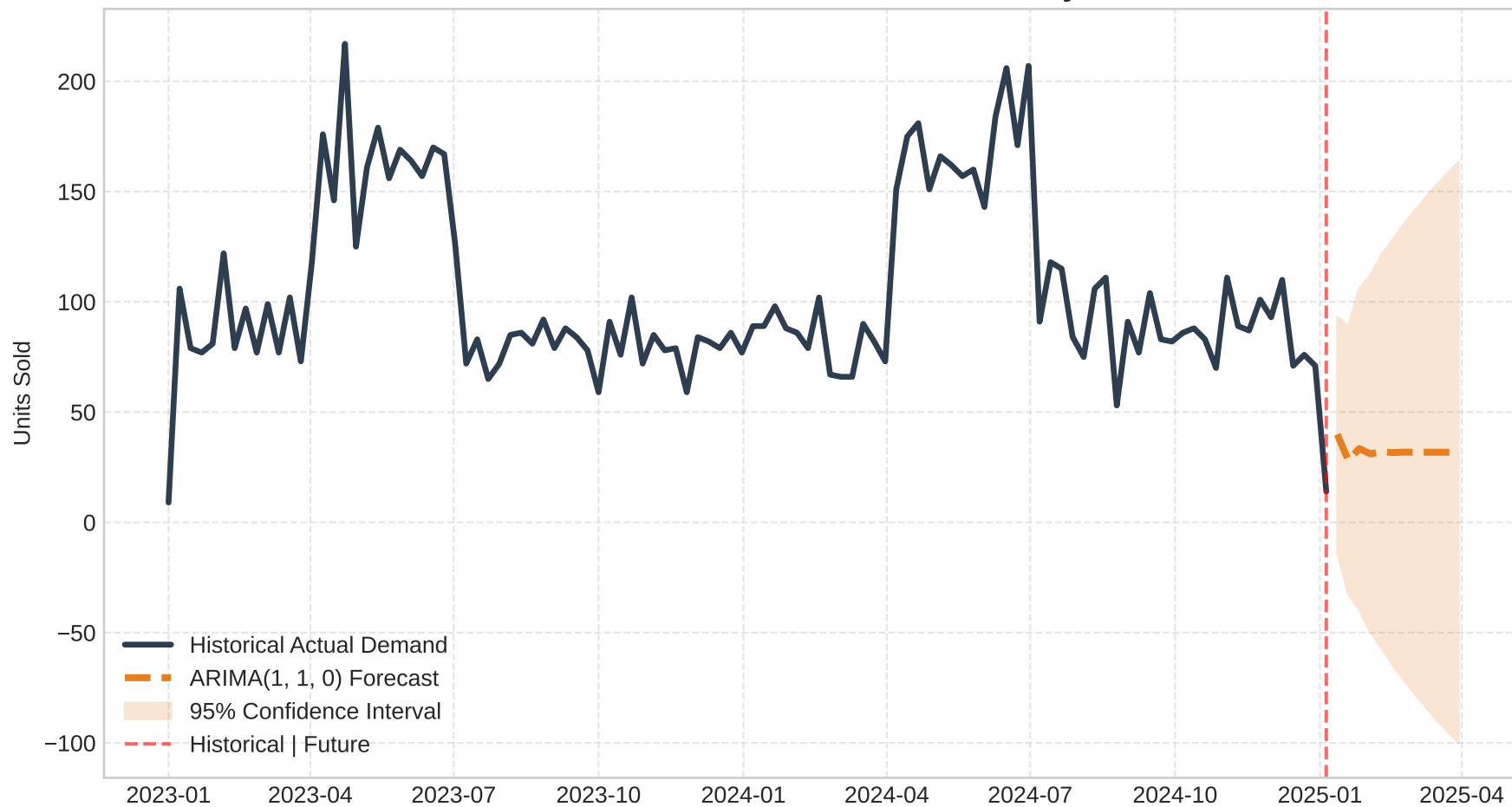
Residual Histogram



Q-Q Plot



Final Production Forecast: 12-Week Weekly Demand



12-Week Predicted Demand Values

Forecast Date	Predicted Demand	Lower CI (95%)	Upper CI (95%)
2025-01-12	39.9	-13.5	93.3
2025-01-19	28.1	-32.7	88.9
2025-01-26	33.5	-39.4	106.3
2025-02-02	31.0	-49.9	112.0
2025-02-09	32.1	-57.0	121.3
2025-02-16	31.6	-64.6	127.9
2025-02-23	31.9	-71.2	135.0
2025-03-02	31.8	-77.6	141.2
2025-03-09	31.8	-83.6	147.2
2025-03-16	31.8	-89.3	152.9
2025-03-23	31.8	-94.7	158.4
2025-03-30	31.8	-100.0	163.6